Human Performance & Limitations · Module A — The Ocean of Air

Altitude & Oxygen

Chapter 3 — The partial pressure of oxygen and the altitude penalty, the four physiological zones of the atmosphere, and the numbers every pilot must know cold.

BOOK	AUTHOR	EXAM	CHAPTER
Human Performance & Limitations	Ghost Aviator	DGCA CPL / ATPL — HPL	3 of 26 · Module A

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1. Partial Pressure of Oxygen — Effects of Increasing Altitude
2. The Physiological Zones of the Atmosphere
3. Self-Check & Memory Aids (Module A)

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Plate 3.0 — Higher is thinner. As the total pressure falls, so does the oxygen your blood can reach — the altitude penalty.

§ 4 Partial Pressure of Oxygen — Effects of Increasing Altitude

As you climb, the **total barometric pressure** drops, and with it the **partial pressure of oxygen** (pO_2). Because the diffusion of oxygen across the alveolar membrane depends on the pressure gradient, what matters to your blood is not the percentage of O_2 — it is the pO_2. Here are the standard values the DGCA expects you to know:

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Standard barometric pressure & oxygen availability at altitude

Altitude	Standard Barometric Pressure	O_2 Available (% of Sea Level)	Pilot Significance
Sea Level (0000 ft)	`101 kPa` (`760 mmHg`)	`100 %`	Normal physiological baseline.
8,000 ft	`75 kPa` (`565 mmHg`)	`74 %`	**Cabin pressure is maintained between 6,000 – 8,000 ft.**
12,000 ft	`64 kPa` (`483 mmHg`)	`64 %`	Lower edge of the Physiological-Deficient Zone.
18,000 ft	`51 kPa` (`380 mmHg`)	`50 %` (≈ half)	**Half the O_2 of sea level.**
24,000 ft	`39 kPa` (`295 mmHg`)	`39 %`	Without supplemental O_2 — incapacitation imminent.
36,000 ft	`23 kPa` (`170 mmHg`)	`22 %` (about a quarter)	**About ¼ of sea-level oxygen (exactly ¼ is near 34,000 ft).**

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THREE DEDUCTIONS YOU MUST BE ABLE TO STATE IN THE EXAM

1. The **availability of oxygen and barometric pressure DECREASE with altitude**.
2. The oxygen available is **about one-fourth** at `36,000 ft` and **half** at `18,000 ft` of the oxygen available at sea level.
3. **Atmospheric pressure drops faster at lower altitudes** in comparison to the same altitude changes at higher altitudes (the pressure–altitude curve is exponential, not linear).

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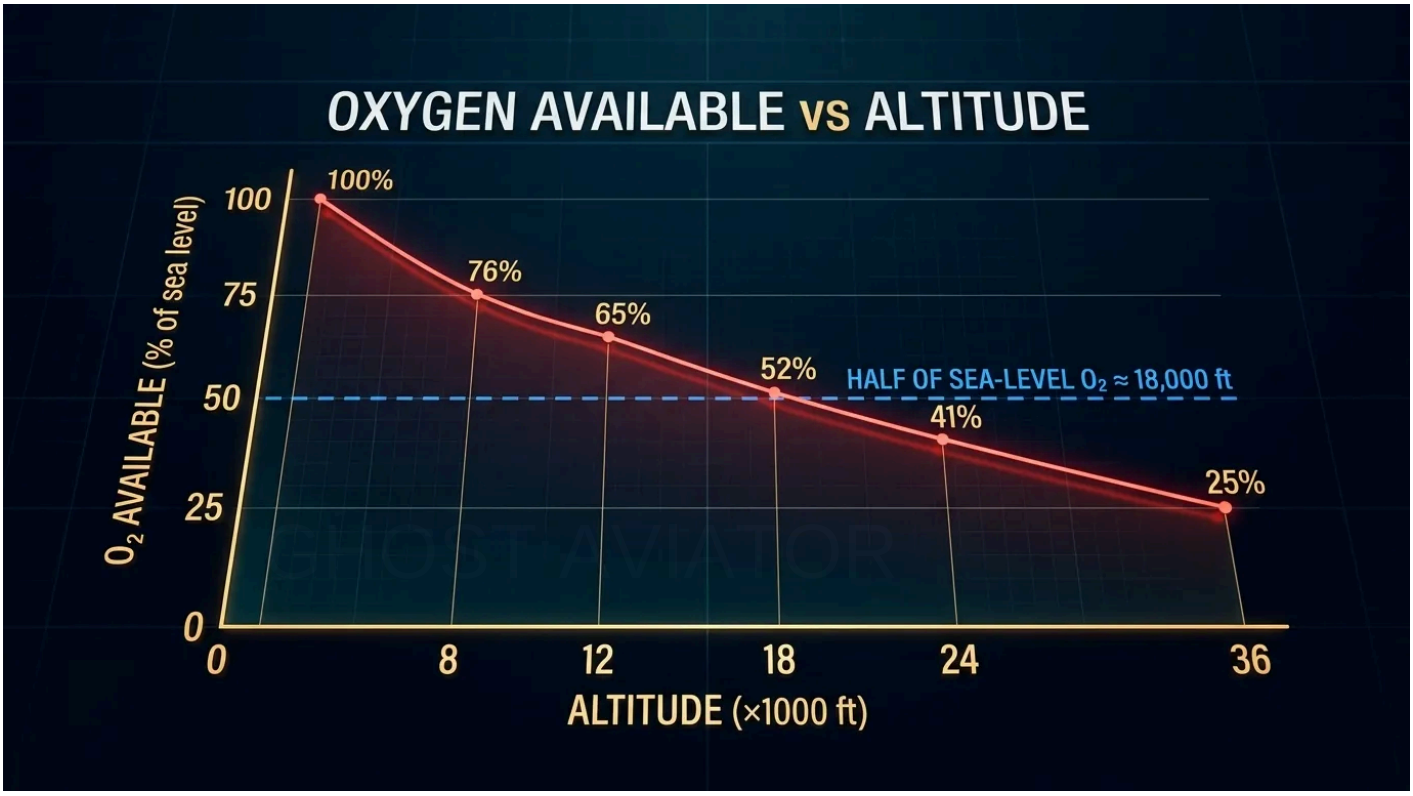


Figure 3.1 — The oxygen available falls non-linearly with altitude: half is gone by 18,000 ft, three-quarters by 36,000 ft.

§ 5 The Physiological Zones of the Atmosphere

The atmosphere is divided — from the pilot's body's point of view — into four functional zones, each demanding a different protective strategy. **You must know the boundaries and the SOP for each.**

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Figure 3.2 — The four physiological zones of the atmosphere and the protection each demands.

5.1 The Physiological Zone

SOP — WITHIN THIS ZONE

Normal healthy human beings used to living near sea level will need **supplementary oxygen to function normally at altitude exceeding** `10,000 to 12,000 ft`. **Pilots will normally begin breathing supplementary oxygen from** `10,000 ft` **above sea level.**

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Figure 3.3 — The oxygen-supply regime by altitude: air only, oxygen+air, 100% oxygen, then oxygen under pressure.

5.2 The Physiological-Deficient Zone

LIMITS & SYMPTOMS — STRICT

Exists from `12,000 ft to 50,000 ft`. **The body is not used to this environment.** The adverse effects include:

- Middle ear and sinus blockage
- Shortness of breath
- Dizziness
- Headache

COMPENSATORY THRESHOLD

At `6,000 - 7,000 ft` altitude the human organism starts with remarkable measures to compensate for the drop in pO_2 when climbing — this is the **threshold for compensatory reactions**. (Hyperventilation, raised heart rate, raised cardiac output, etc. — all later sections.)

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CRITICAL-THRESHOLD LIMITS — NON-PRESSURISED FLIGHT

- A pilot climbing in a **non-pressurised aircraft and WITHOUT using supplemental oxygen** will pass the **"critical threshold" at approximately** `22,000 ft`.
- **Breathing 100 % oxygen** will lift the pilot's physiological safe altitude to approximately `38,000 ft`.

5.3 The Partial Space-Equivalent Zone

DEFINITION & EXTENT

This zone extends from `50,000 ft to 120 nm` (nautical miles). Even 100 % oxygen ceases to be enough at these heights — total pressure becomes the limiting factor.

5.4 The Total Space-Equivalent Zone

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DEFINITION

The Total Space-Equivalent Zone extends **outwards from** `120 nm`.

MANDATORY LIMITS IN THE UPPER TWO ZONES

- **100 % oxygen does NOT protect from hypoxia** — pressure-breathing required.
- **Sealed cabins and pressure suits are a MUST.**
- Blood and body fluids **BOIL above** `63,000 ft` (Armstrong's Line — vapour pressure of body fluids equals ambient pressure).
- **Gravitational changes affect the body** (micro-G adaptation, fluid shift, motion-sickness, etc.).

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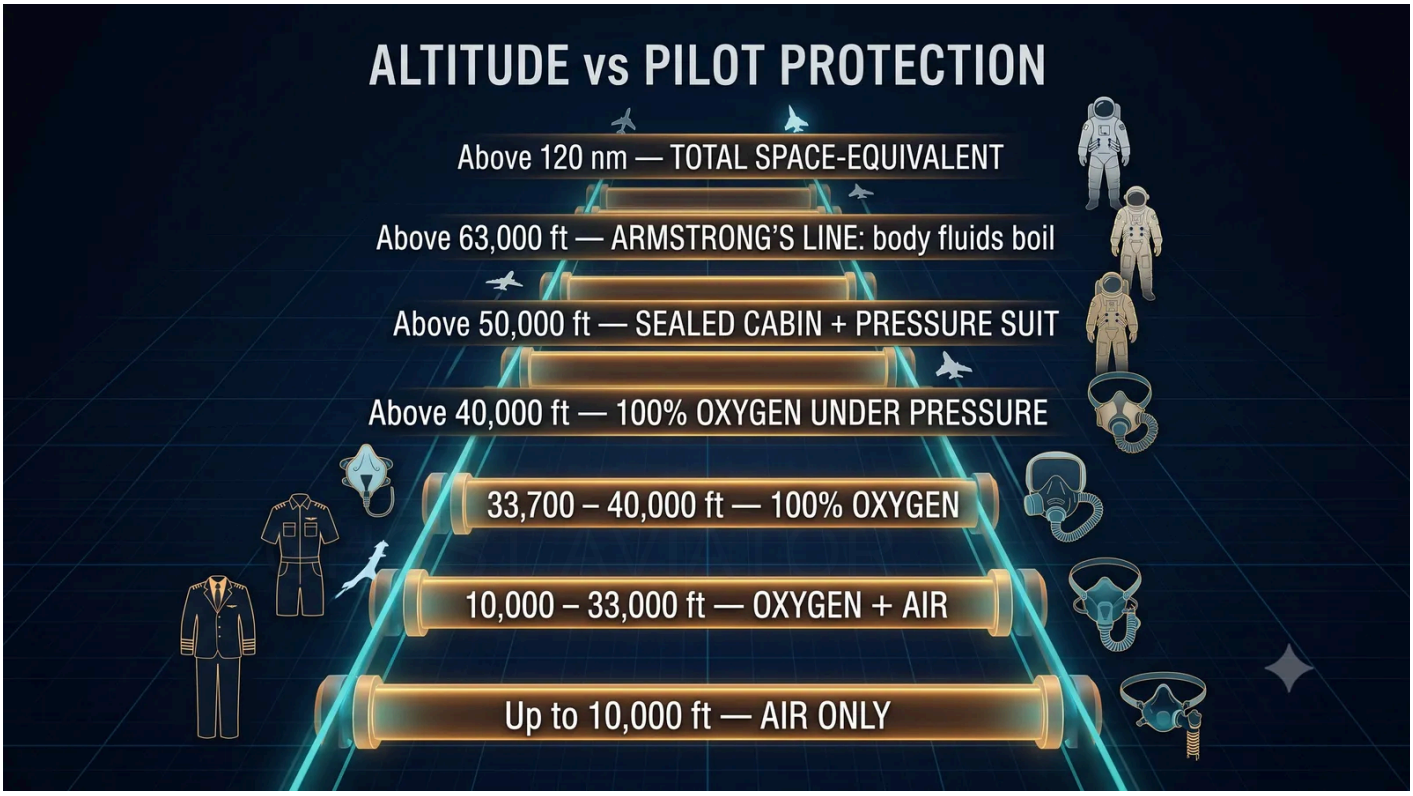


Figure 3.4 — Altitude-versus-protection timeline, from air-only at low level to pressure suits and Armstrong's Line.

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§ 6 Self-Check & Memory Aids

6.1 Numbers you MUST know by heart

Master cheat-sheet — Part 1 numbers

Parameter	Exact Value	Where it comes from
Top of troposphere (average)	`13 km / 8.1 mi / 43,000 ft`	§1 Atmospheric layers
Top of stratosphere (over)	`100,000 ft`	§1
Troposphere holds	`75 %` of atmospheric mass	§1
Nitrogen / Oxygen / Other	`78.084 / 20.95 / ~1 %`	§1.2
Water vapour content	`1 – 4 %` (variable)	§1.2
ISA sea-level pressure	`1013.25 hPa = 760 mmHg = 29.92 inHg = 14.7 psi`	§3.1
ISA lapse rate	`1.98 °C / 1000 ft`	§3.1
Tropopause altitude (ISA)	`36,090 ft`	§3.1
Isothermal temperature	`-56.5 °C`	§3.1
Cabin altitude maintained	`6,000 – 8,000 ft`	§4
O_2 available at 18,000 ft	`52 %` (half of SL)	§4
O_2 available at 36,000 ft	`about 22 %` (roughly one-fourth of SL)	§4
Compensatory reactions begin	`6,000 – 7,000 ft`	§5.2
Pilot starts supplementary O_2	`10,000 ft`	§5.1
Physiological-Deficient Zone	`12,000 – 50,000 ft`	§5.2
Critical threshold (no O_2, unpressurised)	`~22,000 ft`	§5.2
Safe altitude on 100% O_2	`~38,000 ft`	§5.2
100% O_2 Under Pressure (PBA) from	`Above 40,000 ft`	§5.1 fig.
Partial Space-Equivalent Zone	`50,000 ft – 120 nm`	§5.3
Armstrong's Line (body fluids boil)	`63,000 ft`	§5.4
Total Space-Equivalent Zone	`Outwards from 120 nm`	§5.4

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6.2 Common DGCA-style probe questions

TRY ANSWERING THESE WITHOUT LOOKING BACK

1. State the percentage composition of nitrogen and oxygen in the atmosphere. *Does it change with altitude?*
2. Give the ISA sea-level values in **five** different units.
3. What is the ISA lapse rate, and up to what altitude does it apply?
4. Why does hypoxia occur with altitude, when the % of O_2 is unchanged? *(Hint: Dalton's Law.)*
5. At what altitude does a pilot normally start using supplementary oxygen?
6. State the four physiological zones with their altitude boundaries.
7. At what altitude does an unpressurised pilot, NOT using supplemental O_2, hit the critical threshold?
8. Why is a pressure suit mandatory above 50,000 ft? Quote Armstrong's Line value.
9. Explain Boyle's Law and give two flight-medical situations where it applies.
10. How does Henry's Law explain both hypoxia *and* decompression sickness?

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MNEMONIC — 5 GAS LAWS **"Bad Hens Cluck, Geese Dance"** → **B**oyle · **H**enry · **C**harles · **G**raham · **D**alton.

MNEMONIC — ZONES (LOW → HIGH) **P – P-D – PS – TS** → Physiological · Physiological-Deficient · Partial Space-equivalent · Total Space-equivalent. *(Or: "Pilots Probably Prefer Tea" — P, PD, PS, TS.)*

MNEMONIC — HALVING RULE **"Half by Eighteen, Quarter by Thirty-Six"** → Half of sea-level O_2 at **18,000 ft**; one-quarter at **36,000 ft**.

✦ END OF CHAPTER 3 ✦

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